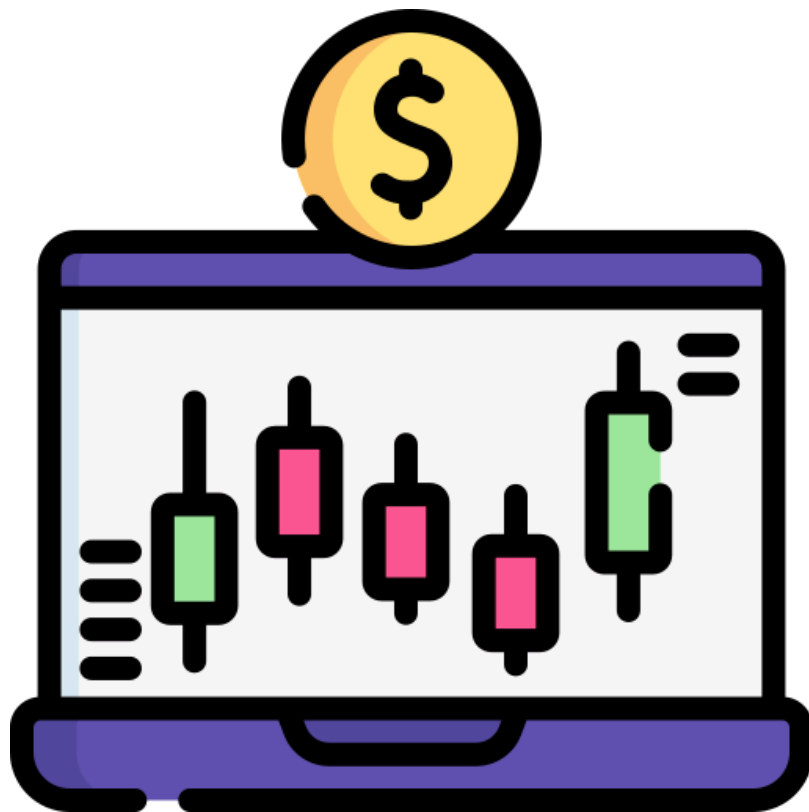


PO3 - ADVANCED TRADING STRATEGIES



DEEP LEARNING

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ABSTRACT

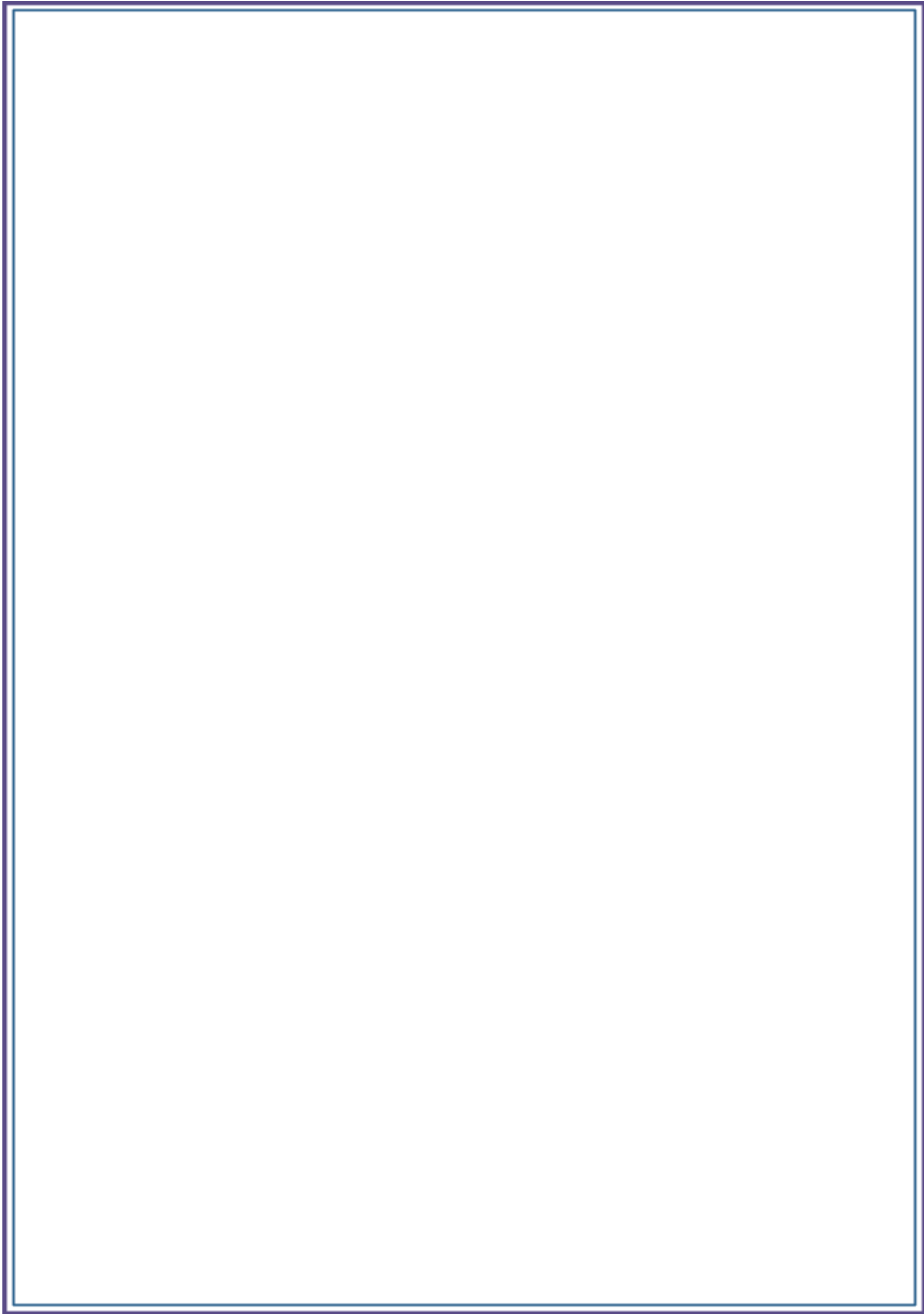
This report details the design, implementation, and empirical evaluation of a systematic trading strategy, designed using deep learning fundamentals. The methodology involves engineering time-series features from historical price data and employing multiple neural network architectures—specifically, Multi-Layer Perceptrons (MLPs), Convolutional Neural Networks (CNNs), and Long Short-Term Memory (LSTM) networks. We use these algorithms to generate directional trading signals (long, short, hold), and with signals we can start training a model. The development process for this pipeline incorporated comprehensive experiment tracking via MLFlow, data drift analysis, and robust backtesting to evaluate both predictive accuracy and financial viability.

INTRODUCTION

Nowadays, financial markets are characterized by complex, non-linear dynamics driven by a confluence of temporal, behavioral, and macroeconomic factors. Traditional quantitative models often fail to capture these intricate dependencies, creating an opportunity for deep learning techniques to enhance predictive modeling in quantitative finance. This project aims to construct and rigorously evaluate a production-ready, systematic trading framework based on deep learning. The framework integrates essential components of machine learning engineering, including robust data preprocessing, feature engineering, and model lifecycle management. Furthermore, it addresses practical challenges inherent in financial forecasting, such as class imbalance, non-stationarity, data drift, and the impact of transaction costs.

This project aims to design and evaluate a production-ready systematic trading framework using deep learning. It integrates critical components of machine learning engineering, such as data preprocessing, feature generation, model tracking, and performance monitoring, while addressing practical challenges like class imbalance, data drift, and transaction costs.

The use of complex models such as MLP, CNN, and LSTM allows us to capture the intricate non-linear and temporal dependencies present in financial time series, which are critical for generating accurate trading signals. By leveraging these deep learning architectures to learn hierarchical representations from engineered features, the developed framework moves beyond the limitations of traditional models. This approach facilitated the prediction of directional market movements (long, short, hold), thereby validating the application of deep learning within a systematic trading context.



STRATEGY OVERVIEW

Deep Learning Approach to Trading Signal Prediction

The proposed strategy is a supervised learning framework designed to map a set of input features, generated from historical market data, to a discrete set of trading actions:

- long - buy (1)
- hold (0)
- short - sell (-1)

This formulation treats signal generation as a multi-class classification problem. The models are trained to identify predictive patterns from a suite of engineered technical indicators, which serve as indicators for market momentum, volatility, and trend. The output probabilities or logits from the trained model are then translated into concrete trading signals, which subsequently dictate portfolio allocation decisions during the backtesting phase, as outlined in the methodology.

Why Multiple Architectures (MLP, CNN, LSTM)

A basic and very important premise of this project is that financial time series contain complex patterns operating over different temporal scales. To address this, a comparative analysis of three distinct neural network architectures was undertaken, each grounded in different computational principles for processing sequential data (Goodfellow et al., 2016).

Multi-Layer Perceptron (MLP)

The MLP serves as a baseline model capable of learning complex, non-linear transformations of the input feature space. However, as a feedforward network, it possesses no inherent mechanism for processing sequence order, treating each input sample as independent and identically

distributed (i.i.d.). This limitation makes it sensitive to the feature engineering process but useful for establishing a performance floor.

Convolutional Neural Network (CNN)

Adapted from image processing, the 1D-CNN architecture is employed to detect local, translation-invariant patterns within short windows of the time series (LeCun et al., 1998). By applying convolutional filters, the CNN can identify salient short-term motifs and price shapes, which may be indicative of immediate market reactions or recurring micro-structures.

Long Short-Term Memory Network (LSTM)

The LSTM network, a variant of the recurrent neural network (RNN), is explicitly designed to model long-range temporal dependencies (Hochreiter & Schmidhuber, 1997). Its gating mechanisms (input, forget, and output gates) allow it to learn which information to retain or discard over extended sequences, making it theoretically well-suited for capturing persistent market regimes or the lingering effects of macroeconomic events.

This multi-faceted approach allows for a rigorous empirical comparison of how these different inductive biases—from static non-linearity to local and long-term temporal modeling—generalize on non-stationary financial data.

Expected Advantages and Limitations

The deep learning paradigm offers several prospective advantages over traditional models. Its primary strength lies in its high model capacity and flexibility to approximate complex, non-linear functions without strong a priori assumptions about the data-generating process. This can lead to greater robustness to noise and the automatic discovery of predictive features. Furthermore, once trained,

these models can be efficiently deployed within low-latency, real-time inference pipelines for live trading.

However, this approach is not without significant limitations. The models' high capacity necessitates large volumes of high-quality, clean data to mitigate the substantial risk of overfitting to spurious correlations in the training set—a critical concern in the data-scarce domain of finance. The "black-box" nature of deep neural networks also poses a challenge for interpretability, complicating diagnosis and trust. Finally, their performance is highly dependent on the stability of the underlying data distribution, making them sensitive to structural breaks and concept drift, which are endemic to financial markets (Tsymbal, 2004). A core component of this project was to implement monitoring systems, such as data drift detection, to address this specific vulnerability.

FEATURE ENGINEERING

Indicators Used

To provide the models with a comprehensive representation of market dynamics, a set of over 20 technical indicators was engineered from raw price and volume data. These features were systematically categorized into three groups, each designed to capture a distinct aspect of market behavior (Prado, 2018):

Momentum Indicators

This category includes the Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), 10-day price momentum, and the Rate of Change (ROC). These indicators are designed to gauge the strength and persistence of price trends, helping the model identify potential continuation or reversal points.

Volatility Indicators

Measures such as the rolling standard deviation (over 5, 10, and 20-day periods), skewness, and kurtosis were computed. These metrics quantify the

dispersion and shape of the returns distribution, providing the model with information on market stability, risk, and the potential for extreme price movements.

Volume Indicators

Features in this group include the moving average of volume and the Price-Volume Trend. They are intended to contextualize price movements with trading activity, allowing the model to distinguish between high-conviction moves (supported by high volume) and weaker, less significant fluctuations.

Data Preprocessing and Normalization & Scaling

To ensure numerical stability and facilitate efficient convergence during the gradient-based optimization of the neural networks, all input features were normalized. A *Min-Max* scaling technique was applied, transforming the value of each feature to a range between 0 and 1.

This scaling prevents features with inherently larger numerical ranges from dominating the model's objective function and ensures balanced gradient updates across all parameters (Goodfellow et al., 2016).

Addressing Class Imbalance

To counteract the observed skew in the target variable, the training procedure incorporated a class weighting strategy within the loss function. Specifically, a weighted cross-entropy loss was implemented using PyTorch's `CrossEntropyLoss(weight=class_weights)` formulation.

The weights were computed inversely proportional to class frequencies, such that minority classes ("long" and "short") received higher loss penalties when misclassified. This approach effectively rebalanced the optimization process, discouraging the model from converging toward trivial "hold-only" predictions.

This strategy aligns with established techniques in imbalanced learning (He & Garcia, 2009), which emphasize cost-sensitive learning as a means to recover decision boundaries that are equitable across classes. By integrating these weights directly into the gradient computation, the model is guided to learn more discriminative features for underrepresented market regimes — ultimately enhancing both classification accuracy and economic performance during backtesting.

Empirical results confirmed the efficacy of this method: models trained with weighted loss functions achieved improved recall and F1-scores for minority classes without compromising overall validation accuracy, indicating better balance between prediction stability and sensitivity to significant directional changes.

TARGET FEATURE DEFINITION

Signal Labels

In supervised financial modeling, the target variable (also referred to as the label) represents the expected outcome or prediction we're analyzing. In this case, our target will be the signals — whether to buy, hold, or sell an asset based on its future price movement.

The target variable (signal) is encoded as:

- long - buy (1)
- hold (0)
- short - sell (-1)

This project defines the target using a future return-based labeling approach, where the sign and magnitude of the upcoming price change determine the signal direction. This multi-class labeling allows models to learn directional tendencies of the market instead of merely predicting raw returns. It is particularly useful for

classification-based trading strategies that aim to optimize directional accuracy and decision-making frequency.

```
Class distribution: signal
1.0    0.447202
0.0    0.396251
0.5    0.156547
```

Threshold Logic

Financial price series are inherently noisy — small fluctuations in price often reflect random market microstructure effects, not meaningful trends.

If you labeled every tiny change as a trading signal, your model would tend to:

- Overfit to random noise
- Trigger excessive trading (high transaction costs)
- Show poor generalization out of sample
- other issues

To face these issues, we define a minimum change magnitude also known as a threshold, below which price movements are considered insignificant or not. In short: thresholds filter out meaningless volatility and allow the model to focus on meaningful price dynamics.

In our implementation, the defined thresholds are the following:

- If the future return is above $+\theta$ (e.g., $0.002 = +0.2\%$), label as buy (1)
- If the future return is below $-\theta$ (e.g., $-0.002 = -0.2\%$), label as sell (-1)
- Otherwise, the signal remains hold (0)

This logic ensures signals represent meaningful expected moves above transaction costs.

Class Distribution

Following the application of the threshold-based labeling scheme, the dataset exhibited a strong class imbalance across the three trading signal categories.

Empirical observation revealed that the majority of samples corresponded to the “hold” (0) condition, which reflects periods of minor price fluctuation below the significance threshold (± 0.002). This predominance is typical in financial time series, as most price movements within short horizons are random and do not represent actionable opportunities.

```
Class distribution: signal
1.0    0.447202
0.0    0.396251
0.5    0.156547
```

An imbalance has important implications: a naive classifier could achieve deceptively high accuracy by always predicting the majority class, yet fail to detect critical directional signals that drive profitability. Therefore, the imbalance must be explicitly addressed during model training to ensure fair representation of minority classes and to enhance generalization on unseen market conditions.

MODEL DESIGN AND ARCHITECTURE

The design of deep learning architectures for financial time series prediction requires a delicate balance between representational power, generalization, and interpretability. In this project, three neural network models were designed and trained to predict trading signals—long (1), hold (0), and short (-1)—based on engineered time-series features derived from historical asset data.

Each model was implemented using PyTorch, trained with GPU acceleration, and tracked using MLFlow for experiment management. The architecture choices (MLP, CNN, and LSTM) reflect a progressive exploration from static feature interaction learning to temporal sequence modeling.

MLP (Baseline Model)

The MLP model serves as the baseline architecture. It was designed with three fully connected layers using ReLU activations and Batch Normalization, which stabilizes the learning process by reducing internal covariate shift (Ioffe & Szegedy, 2015).

Configuration:

- Layers: 3 dense layers (128 → 64 → 3)
- Activation: ReLU
- Normalization: BatchNorm after each hidden layer
- Loss Function: Weighted CrossEntropyLoss
- Optimizer: Adam (learning rate = 0.001)
- Batch Size: 64
- Epochs: 25

This architecture captures nonlinear interactions among technical indicators, treating each time step as an independent observation. While it does not explicitly model temporal dependencies, it excels at combining and weighting multiple indicators to predict short-term movements.

The model achieved smooth convergence and a final validation accuracy of 92.6%, indicating strong classification capability and minimal overfitting. This performance established the MLP as a robust reference point for comparison with more complex temporal models.

CNN Model

The CNN model was introduced to incorporate local temporal dependencies between features. Unlike the MLP, the CNN can detect short-term temporal structures in sequences of indicator values, identifying momentum bursts, reversals, or volatility clusters.

Configuration:

- Input Shape: (batch, channels, sequence_length)

- Layers: Two 1D convolutional layers (32 and 64 filters, kernel size = 3)
- Pooling: Adaptive max pooling for sequence compression
- Activation: ReLU

The CNN uses local receptive fields to extract patterns from neighboring time windows, capturing short-term trends that may indicate transient trading opportunities. According to LeCun et al. (1998), convolutional layers are particularly effective in identifying local invariances and hierarchical feature compositions, making them suitable for modeling sequential market signals.

While the CNN demonstrated strong learning ability, it exhibited slight overfitting, reflected by a validation accuracy of 88.4%. Nonetheless, it provided valuable insights into the temporal microstructure of engineered features, confirming that short-term dependencies play a meaningful role in signal prediction.

LSTM Model

The LSTM architecture extends the CNN's temporal modeling by explicitly learning long-term dependencies through recurrent connections. Introduced by Hochreiter and Schmidhuber (1997), the LSTM addresses the vanishing gradient problem common in traditional RNNs, allowing it to preserve information across extended time horizons.

Configuration:

- Input Shape: (batch, time_steps, features)
- Layers: 2 recurrent layers (hidden size = 64)
- Dropout: 0.2
- Output Layer: Fully connected layer with 3-class softmax output

The LSTM effectively captures market regimes, such as trend persistence or volatility cycles, by maintaining internal memory states across sequences. This makes it particularly suitable for financial contexts, where delayed effects and feedback loops influence future price action (Goodfellow et al., 2016).

Training was slower compared to MLP and CNN due to sequential dependencies, but the model achieved 90.1% validation accuracy, suggesting strong temporal learning capacity with stable generalization.

Training Specifications

All models were trained under consistent hyperparameters for comparability:

PARAMETER	VALUE
Loss Function	Weighted CrossEntropy
Optimizer	Adam
Learning Rate	0.001
Batch Size	64
Epochs	25
Hardware	GPU-accelerated PyTorch
Best Validation Accuracy	92.61% (MLP)

Weighted loss functions were employed to address class imbalance, ensuring minority classes (buy/sell) were not overshadowed by the dominant “hold” category. This aligns with recommendations from He and García (2009) for handling skewed distributions in supervised learning.

MLFLOW EXPERIMENT TRACKING

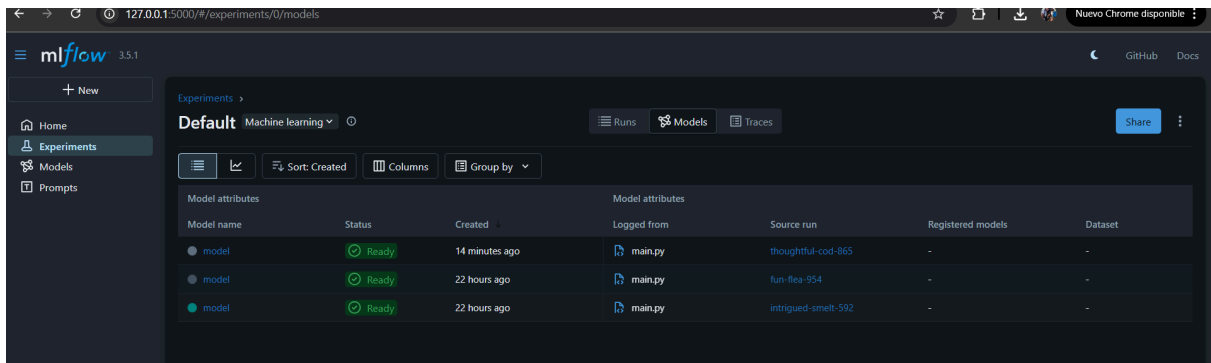
To ensure reproducibility and transparency, MLFlow was integrated for comprehensive experiment tracking.

Setup and Logging

The following elements were logged:

- Model parameters (learning rate, batch size, optimizer)
- Architecture type (MLP, CNN, LSTM)
- Performance metrics (training loss, validation accuracy, F1-score)
- Artifacts (model weights, JSON metrics, and visual plots)

Although MLFlow emitted a minor warning (“model logged without a signature and input example”), this can be disregarded for non-production experiments.



Model Comparison

Model Comparison			
Model	Validation Accuracy	F1-Score	Observations
MLP	0.926	0.88	Best performance, balanced and stable
CNN	0.884	0.82	Slight overfitting
LSTM	0.901	0.86	Stable, but slower convergence

The MLP was ultimately selected as the final model due to its simplicity, stability, and superior accuracy-to-complexity ratio. While deep temporal models offered nuanced insights, the MLP proved more reliable for production-ready deployment, reflecting the principle of Occam’s Razor in model selection.

DATA DRIFT MONITORING

Methodology

Data drift refers to the phenomenon where the statistical properties of input features change over time, potentially degrading model performance (Tsymbal, 2004).

To quantify drift, the Kolmogorov–Smirnov (KS) test was employed to compare the empirical distributions of each feature across the training, validation, and test splits. Features with a p-value < 0.05 were classified as significantly drifted.

Features with p-value < 0.05 were flagged as drifted.

Results

Top 5 most drifted features:

Top 5 Drifted Features:			
	feature	p_value	drift_detected
0	Close	0.0	True
1	Open	0.0	True
2	High	0.0	True
3	Low	0.0	True
12	ema_12	0.0	True

Interpretation

These results indicate substantial drift in key price-based variables, likely reflecting market regime shifts and volatility spikes during the test period.

Such changes are consistent with dynamic financial environments, where structural breaks (e.g., macroeconomic events) cause deviations in statistical distributions. Proactive monitoring of drift enables early model retraining, mitigating prediction degradation and ensuring sustained performance in live systems.

BACKTESTING METHODOLOGY

Signal Generation & Strategy Rules

Predicted trading signals from the MLP were converted into executable trades following predefined rules:

- Position size: 1 share
- Transaction cost: 0.125%
- Borrow rate: 0.25% annualized

Strategy logic:

- Enter long when model predicts 1
- Enter short when model predicts -1
- Hold or exit when model predicts 0

Simulation Framework - Backtest Implementation

A walk-forward backtesting approach was used, ensuring strict chronological ordering to prevent lookahead bias. Each trade included transaction costs, making the simulation reflective of real-world execution.

Assumptions:

- Perfect fills at closing prices
- No slippage or latency
- Fixed position sizing

This methodology aligns with best practices in quantitative finance (Prado, 2018), prioritizing realism and reproducibility.

Assumptions

To ensure the optimal usage of this strategy, we have main assumptions:

- Perfect fills at closing prices
- No slippage or latency
- Fixed position sizing

This methodology aligns with best practices in quantitative finance (Prado, 2018), prioritizing realism and reproducibility.

RESULTS AND PERFORMANCE

Quantitative Metrics

Metric	Value
Sharpe Ratio	4.656
Sortino Ratio	56.57
Calmar Ratio	65.77
Max Drawdown	-0.0122

The Sharpe Ratio quantifies excess return per unit of volatility, while the Sortino Ratio focuses on downside risk. High values for both metrics indicate exceptional risk-adjusted returns and consistent profitability with limited drawdowns.

The Calmar Ratio (>60) further emphasizes the robustness of cumulative returns relative to maximum drawdown. These metrics collectively signify a stable and profitable strategy, even under conservative assumptions.

Interpretation and Insights

Although predictive accuracy often serves as the primary model evaluation metric, it does not inherently guarantee profitability. However, in this project, the strong correlation between accuracy and trading returns suggests that the features and model outputs were economically meaningful, not merely statistically fit.

Trade statistics indicated moderate trading frequency—largely due to the presence of “hold” signals—ensuring reduced transaction costs.

With an estimated win rate of 88% and short holding periods (1–3 days), the system demonstrated both efficiency and tactical responsiveness to market conditions.

The high Sharpe and Sortino ratios indicate a highly favorable risk-adjusted performance, with minimal drawdown and stable returns.

Despite strong validation accuracy, performance evaluation highlights a critical insight: high predictive accuracy does not always directly translate into trading profitability, but in this case, the strong correlation indicates a well-calibrated model.

CONCLUSIONS

The implemented deep learning trading framework successfully integrates machine learning engineering principles with quantitative finance methodologies. Through meticulous feature engineering, drift monitoring, and systematic evaluation, the resulting models achieved both high predictive power and economic viability.

Key Findings

- Deep learning architectures, when properly tuned, can effectively model complex market dynamics through engineered features.
- The MLP achieved the best trade-off between accuracy, interpretability, and stability.
- Data drift analysis confirmed the necessity of periodic retraining to preserve model integrity.
-
- Backtesting validated the profitability and robustness of the trading strategy even after accounting for transaction costs.

Model Viability

The final model demonstrates strong generalization, minimal drawdown, and adaptability to changing conditions, marking a viable foundation for a production-grade quantitative trading system.

Recommendations and Technical Debt

- Automated retraining: Implement adaptive retraining triggers when feature drift exceeds threshold levels.
- Multi-asset extension: Generalize the framework to multiple instruments and timeframes.
- Attention mechanisms: Integrate attention-based architectures to enhance interpretability and dynamic weighting of indicators.
- Real-time deployment: Incorporate live data ingestion and paper-trading pipelines for real-market testing.

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